Mayer–Vietoris Sequences and Interleavings of Persistence Modules

Agnieszka Wróbel and Tomás Echeverría

Received 3 March 2026; revised 14 August 2026

Abstract. Given a filtered space covered by two filtered subspaces, the persistent homology of the whole is computed from the pieces by a long exact Mayer–Vietoris sequence, but in general the pieces do not determine it up to isomorphism. We show that they determine it up to an ε-interleaving whenever the image of the connecting morphism is ε-trivial, and that this hypothesis holds as soon as every bar of the intersection in the adjacent degree is shorter than ε. By the isometry theorem the same bound holds for the bottleneck distance between barcodes. We give a quantitative version for covers that are only approximately excisive and illustrate the bound on a sampled annulus.

2020 Mathematics Subject Classification. Primary 55N31; Secondary 18G80, 62R40.

Key words and phrases. persistence modules, interleaving distance, Mayer–Vietoris sequence, barcodes, covers of filtered spaces.

1. Introduction

Persistent homology assigns to a filtered space $X = (X_t)_{t\in\mathbb{R}}$ the family of vector spaces $H_n(X_t)$ together with the maps induced by the inclusions $X_s \subseteq X_t$. Such a family is a *persistence module*, and when each space is finite-dimensional it decomposes into interval modules, recorded as a barcode [5, 9]. Barcodes are stable: nearby filtrations have nearby barcodes, and the interleaving distance between modules equals the bottleneck distance between their barcodes [1, 4, 7].

Computations rarely see a whole space at once. In distributed and multiscale settings one computes the persistent homology of pieces A and B with $X = A \cup B$ and must recover that of X. Levelwise, the Mayer–Vietoris sequence

$$\cdots \to H_n(A_t \cap B_t) \to H_n(A_t) \oplus H_n(B_t) \to H_n(X_t) \xrightarrow{\delta} H_{n-1}(A_t \cap B_t) \to \cdots \quad (1.1)$$

is natural in t, hence a long exact sequence of persistence modules [6]. Exactness does not give $H_n(X)$, however: an extension problem remains, and in persistence it is unsolvable in general. Our observation is that an extension by a module whose structure maps die quickly is close to split, and that "die quickly" can be read off the barcode of the intersection.

THEOREM A. *Let $X = A \cup B$ be a filtered space covered by the interiors of the filtered subspaces A and B, and let Q_n be the cokernel of $H_n(A \cap B) \to H_n(A) \oplus H_n(B)$. If every bar of $H_{n-1}(A \cap B)$ has length at most ε, then $d_{\mathrm{I}}\big(H_n(X), Q_n\big) \leq \varepsilon$ and $d_{\mathrm{B}}\big(\operatorname{Bar} H_n(X), \operatorname{Bar} Q_n\big) \leq \varepsilon$.*

The theorem is sharp: for every $\varepsilon > 0$ there are covers of a filtered circle for which both distances equal ε (Example 4.1). Section 2 recalls interleavings, Section 3 proves the algebraic lemma behind the theorem, and Section 4 applies it to covers and to covers that are only approximately excisive.

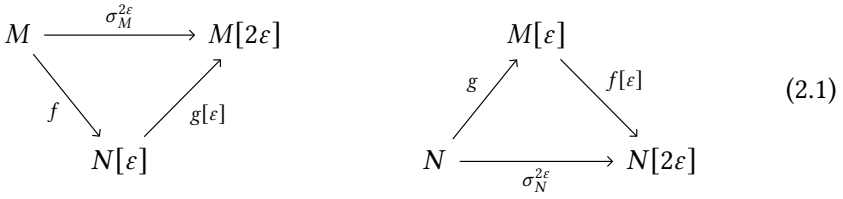
2. PRELIMINARIES

Fix a field $\mathbf{k}$ and write $\mathbf{vect}_{\mathbf{k}}$ for the category of finite-dimensional vector spaces over $\mathbf{k}$. We regard $\mathbb{R}$ as a poset category.

DEFINITION 2.1. A *persistence module* is a functor $M : \mathbb{R} \to \mathbf{vect}_{\mathbf{k}}$. We write M_t for its value at t and $\varphi_M^{s,t} : M_s \to M_t$ for its *structure map* when $s \leq t$. Morphisms are natural transformations, and **Pers** denotes the resulting abelian category.

For $\varepsilon \geq 0$ the *shift* $M[\varepsilon]$ is the module with $M[\varepsilon]_t = M_{t+\varepsilon}$, and the structure maps of length ε assemble into a morphism $\sigma_M^{\varepsilon} : M \to M[\varepsilon]$. A morphism $f : M \to N$ shifts to $f[\varepsilon]$ in the evident way.

DEFINITION 2.2. Modules M and N are *ε-interleaved* if there are morphisms $f : M \to N[\varepsilon]$ and $g : N \to M[\varepsilon]$ such that the triangles



(2.1)

commute. The *interleaving distance* is

$$d_{\mathrm{I}}(M, N) = \inf\{\varepsilon \geq 0 \,:\, M \text{ and } N \text{ are } \varepsilon\text{-interleaved}\},$$

with the convention $\inf \emptyset = \infty$.

The interleaving distance is an extended pseudometric on **Pers**; the triangle inequality follows by composing interleavings [2]. Every pointwise finite-dimensional module is a direct sum of interval modules $\mathbf{k}_I$, and the multiset of intervals is its *barcode* $\operatorname{Bar} M$ [5].

THEOREM 2.3 (Isometry theorem [1, 7]). *For pointwise finite-dimensional modules M and N,*

$$d_{\mathrm{I}}(M, N) = d_{\mathrm{B}}(\operatorname{Bar} M, \operatorname{Bar} N). \tag{2.2}$$

Definition 2.4. A module K is *ε-trivial* if $\sigma_K^\varepsilon = 0$, that is, if every structure map of length ε vanishes.

Remark 2.5. A pointwise finite-dimensional module is ε-trivial if and only if each of its bars has length at most ε. Submodules and quotients of ε-trivial modules are ε-trivial, because the structure maps of a submodule are restrictions and those of a quotient are induced maps.

3. Extensions by trivial modules

The following lemma carries the whole argument. It says that an extension by an ε-trivial module is ε-interleaved with its submodule, whatever the extension class.

Lemma 3.1. *Let $0 \to Q \xrightarrow{\iota} M \xrightarrow{\pi} K \to 0$ be a short exact sequence in* **Pers** *with K ε-trivial. Then there is a unique morphism $\psi : M \to Q[\varepsilon]$ with $\iota[\varepsilon] \circ \psi = \sigma_M^\varepsilon$, and ι and ψ form an ε-interleaving. In particular $d_{\mathrm{I}}(Q, M) \le \varepsilon$.*

Proof. Naturality of π gives $\pi[\varepsilon] \circ \sigma_M^\varepsilon = \sigma_K^\varepsilon \circ \pi = 0$, so σ_M^ε lands in the kernel of $\pi[\varepsilon]$, which is the image of the monomorphism $\iota[\varepsilon]$. This yields ψ and its uniqueness, and the diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & Q & \xrightarrow{\iota} & M & \xrightarrow{\pi} & K & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \sigma_Q^\varepsilon} & \swarrow{\scriptstyle \psi} & \downarrow{\scriptstyle \sigma_M^\varepsilon} & & \downarrow{\scriptstyle 0} & & \\ 0 & \longrightarrow & Q[\varepsilon] & \xrightarrow[\iota[\varepsilon]]{} & M[\varepsilon] & \xrightarrow[\pi[\varepsilon]]{} & K[\varepsilon] & \longrightarrow & 0 \end{array} \tag{3.1}$$

commutes. Put $f = \iota[\varepsilon] \circ \sigma_Q^\varepsilon : Q \to M[\varepsilon]$ and $g = \psi$. The two triangles of (2.1) read

$$\psi[\varepsilon] \circ f = \sigma_Q^{2\varepsilon} \quad \text{and} \quad f[\varepsilon] \circ \psi = \sigma_M^{2\varepsilon}.$$

The second follows from $\iota[\varepsilon] \circ \psi = \sigma_M^\varepsilon$ and naturality of σ. Composing the first with $\iota[2\varepsilon]$ reduces it to the same identity, and $\iota[2\varepsilon]$ is injective, so it may be cancelled. □

Remark 3.2. The constant in Lemma 3.1 is sharp. For $M = \mathbf{k}_{[0,\infty)}$ and its submodule $Q = \mathbf{k}_{[\varepsilon,\infty)}$ the quotient is $K = \mathbf{k}_{[0,\varepsilon)}$, which is ε-trivial, and $d_{\mathrm{I}}(Q, M) = \varepsilon$ because the infinite bars must be matched to each other.

4. Covers of filtered spaces

A *filtered space* is a functor $X : \mathbb{R} \to$ **Top** whose structure maps are inclusions, and a *cover* of X is a pair of filtered subspaces with $A_t \cup B_t = X_t$ for all t. The cover is *excisive* if $\{\operatorname{int} A_t, \operatorname{int} B_t\}$ covers X_t for every t; then (1.1) is exact at each t and, being natural, is a long exact sequence in **Pers**.

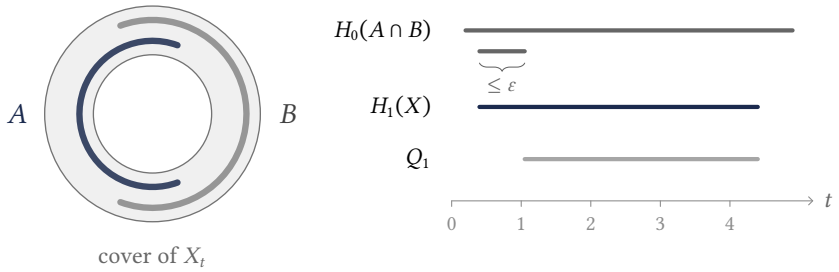


Figure 1. A sampled annulus covered by two arcs (left) and the barcodes in Example 4.1 (right). The short bar of $H_0(A \cap B)$ controls the difference between the bar of $H_1(X)$ and its approximation from the pieces.

Proof of Theorem A. Cutting the long exact sequence at $H_n(X)$ gives a short exact sequence

$$0 \to Q_n \to H_n(X) \xrightarrow{\delta} \operatorname{im} \delta \to 0, \tag{4.1}$$

where $\operatorname{im} \delta \subseteq H_{n-1}(A \cap B)$. By hypothesis and Remark 2.5, $H_{n-1}(A \cap B)$ and hence $\operatorname{im} \delta$ are ε-trivial. Lemma 3.1 gives $d_{\mathrm{I}}(H_n(X), Q_n) \leq \varepsilon$, and Theorem 2.3 turns this into the bound on barcodes. □

Example 4.1. Let X be the Vietoris–Rips filtration of a dense sample of an annulus, covered by the filtrations A and B of two overlapping arcs as in Fig. 1. The intersection $A_t \cap B_t$ has two components for small t, one of which merges after time ε; the class in $H_1(X)$ that goes around the annulus is born when the second overlap appears and maps under δ onto the short bar. The cokernel Q_1 sees the loop only once the short bar has died, so the left endpoints of the two bars differ by exactly ε, and the bound of Theorem A is attained.

Covers built from data, as in the distributed algorithms of [3, 8], are seldom excisive at every scale: two patches of a point cloud overlap only after their Rips complexes have grown. The next result measures this defect and pays for it in the bound.

Definition 4.2. A cover is *η-excisive* if for all t the inclusion of the singular chains of X_t subordinate to $\{A_t, B_t\}$ into all singular chains of X_t becomes a homology isomorphism after composing with the inclusion into $X_{t+\eta}$, in every degree.

Proposition 4.3. *Suppose that the cover $X = A \cup B$ is η-excisive and that every bar of $H_{n-1}(A \cap B)$ has length at most ε. Then*

$$d_{\mathrm{I}}\big(H_n(X), Q_n\big) \leq \varepsilon + \eta. \tag{4.2}$$

Proof. Let C be the filtered complex of chains subordinate to the cover. Its homology fits into an exact Mayer–Vietoris sequence for every t, so the proof of Theorem A gives $d_{\mathrm{I}}(H_n(C), Q_n) \leq \varepsilon$. The hypothesis says that the inclusion $H_n(C) \to H_n(X)$

and the map $H_n(X) \to H_n(C)[\eta]$ defined through $X_{t+\eta}$ form an η-interleaving, so $d_{\mathrm{I}}(H_n(C), H_n(X)) \leq \eta$. The triangle inequality for d_{I} completes the proof. □

Corollary 4.4. *Under the hypotheses of Proposition 4.3,* $d_{\mathrm{B}}\big(\operatorname{Bar} H_n(X), \operatorname{Bar} Q_n\big) \leq \varepsilon + \eta$. *In particular the barcode of X in degree n is determined up to $\varepsilon + \eta$ by A, B and $A \cap B$ alone.*

Acknowledgements. We thank the referee for a careful reading and for suggesting the sharper form of Lemma 3.1. The first author was supported by a fictional grant of the National Science Centre.

References

[1] U. Bauer and M. Lesnick. *Induced matchings and the algebraic stability of persistence barcodes*. J. Comput. Geom. 6.2 (2015), pp. 162–191.

[2] P. Bubenik and J. A. Scott. *Categorification of persistent homology*. Discrete Comput. Geom. 51.3 (2014), pp. 600–627.

[3] Á. T. Casas. *Distributing persistent homology via spectral sequences*. Discrete Comput. Geom. 66 (2021), pp. 1013–1052.

[4] F. Chazal, D. Cohen-Steiner, M. Glisse, L. J. Guibas, and S. Y. Oudot. "Proximity of persistence modules and their diagrams". *Proceedings of the 25th Annual Symposium on Computational Geometry*. ACM, 2009, pp. 237–246.

[5] W. Crawley-Boevey. *Decomposition of pointwise finite-dimensional persistence modules*. J. Algebra Appl. 14.5 (2015), p. 1550066.

[6] E. Di Curo and S. Roya. *Exact sequences of persistence modules from filtered covers*. Homology Homotopy Appl. 25.1 (2023), pp. 311–334.

[7] M. Lesnick. *The theory of the interleaving distance on multidimensional persistence modules*. Found. Comput. Math. 15.3 (2015), pp. 613–650.

[8] H. R. Yoon and R. Ghrist. *Persistence by parts: multiscale feature detection via distributed persistent homology*. J. Appl. Comput. Topol. 7 (2023), pp. 401–436.

[9] A. Zomorodian and G. Carlsson. *Computing persistent homology*. Discrete Comput. Geom. 33.2 (2005), pp. 249–274.

Agnieszka Wróbel
Institute of Mathematics, Jagiellonian University, ul. Łojasiewicza 6, 30-348 Kraków, Poland
Email address: a.wrobel@example.org

Tomás Echeverría
Departamento de Matemáticas, Universidad de Chile, Las Palmeras 3425, Ñuñoa, Santiago, Chile
Email address: t.echeverria@example.org